

Hardik Santosh Gaonkar

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AI & ML undergraduate building AI systems end to end — autonomous agents, RAG, ML and computer vision — and proving they work with held-out benchmarks, confidence intervals and CI quality gates. Seeking AI/ML engineering internships.

EDUCATION

BMS Institute of Technology and Management, Bengaluru 2024 – 2028 (expected)
B.E. in Artificial Intelligence and Machine Learning CGPA: 8.09 / 10
Class XII (PUC), **St. Joseph College of Commerce and Science: 93.6%** · Class X, **Balmandir High School: 97.6%**

PROJECTS

AegisOps | Autonomous incident-response agent for Kubernetes (AIOps) github.com/hrdk6/AegisOps
Go · controller-runtime · Python · FastAPI · PostgreSQL · Kubernetes · OpenTelemetry · Prometheus · Next.js

- Built a closed-loop AIOps agent that detects SLO violations, turns metrics, logs, traces and deploy history into cited evidence, diagnoses root cause with calibrated confidence, fixes it and verifies recovery.
- Enforced bounded autonomy with a deterministic Go policy engine (risk scoring, budgets, cooldowns, circuit breaker, stale-plan checks), HMAC-signed human approvals and digital-twin sandbox tests of risky fixes.
- Routed optional LLM diagnosis across 5 providers (OpenAI, Gemini, Groq, NVIDIA, Ollama) with fallback and structured output; model output must cite stored evidence and may only request 12 typed actions — no shell.
- Benchmarked 21 injected failures in a live cluster against hidden ground truth: pass rate rose from **76% to 100%** over three runs as defects were fixed, with **0 unsafe actions** and 140s median recovery.

CareFlow AI | Clinical AI platform with access-controlled RAG, ML and X-ray triage github.com/hrdk6/CareFlow-AI
Python · FastAPI · PostgreSQL + pgvector · scikit-learn · XGBoost · SHAP · ONNX Runtime · Next.js · TypeScript · Docker

- Built a full-stack clinical AI platform where the LLM sees only data the user may access (RBAC + row-level SQL filters), and a 3-pass PII gateway (vault, regex, BERT NER) masks patient identities before cloud LLM calls.
- Implemented hybrid RAG (bge-small embeddings + BM25, RRF fusion, cross-encoder reranking, citation validation), reaching **Recall@1 0.955 and MRR 0.977** on a 48-question benchmark; reranking lifted context precision from 0.62 to 0.84.
- Trained explainable risk models on 99K patient encounters: a calibrated Random Forest for 30-day readmission (**PR-AUC 2× base rate**, SHAP) and XGBoost length of stay, with patient-grouped splits and leakage ablations.
- Applied transfer learning to chest X-ray triage: logistic heads on frozen ResNet-50 (ONNX) features, trained on 26K NIH ChestX-ray14 films; **8 findings reached ROC-AUC 0.71–0.80** on 7.5K patient-disjoint test films, with class activation maps.

PatchPilot | Autonomous bug-fixing coding agent with sandboxed execution github.com/hrdk6/PatchPilot
Python · LangGraph · FastAPI · SQLAlchemy · Qdrant · Docker · React · TypeScript · GitHub Actions

- Developed an agent that turns a repository and a bug report into a tested patch: a 9-node **LangGraph** state machine (index → retrieve → plan → patch → sandbox test → repair) whose budget-capped repair loop feeds test failures back to the LLM.
- Designed explainable code retrieval: AST-based, symbol-level chunking and hybrid ranking over **7 weighted signals** (semantic, lexical, import/call graph, failure trace, ...), logging why each chunk entered the prompt.
- Sandboxed LLM-written code in hardened Docker (no network, non-root, capabilities dropped, resource limits) behind a patch policy; added a multi-model benchmark harness with Wilson CIs, API-key auth and **537 tests**.

GroundTruth | Self-verifying RAG with a rigorous evaluation harness github.com/hrdk6/GroundTruth
Python · FastAPI · PostgreSQL + pgvector · Sentence-Transformers · Claude / NVIDIA NIM · Next.js

- Engineered version-aware RAG over Kubernetes docs (pgvector HNSW + BM25 written in SQL, RRF fusion) that checks every generated sentence against its cited source and regenerates or abstains: **faithfulness 0.94–0.97**.
- Evaluated every change with bootstrap 95% CIs, paired significance tests and a CI regression gate; a self-audit caught **13 measurement bugs**, including one that produced a spurious +0.50 recall@5 gain.

TECHNICAL SKILLS

Languages	Python, Go, TypeScript, JavaScript, SQL
GenAI & LLMs	RAG, hybrid search (BM25 + dense), reranking, embeddings, LLM agents, LangGraph, tool calling, prompt engineering, LLM evaluation, guardrails; Claude, OpenAI, Gemini, Groq, NVIDIA NIM, Ollama
ML, NLP & CV	scikit-learn, XGBoost, SHAP, pandas, NumPy, Hugging Face, ONNX Runtime, NER, transfer learning
Backend & Web	FastAPI, Pydantic, SQLAlchemy, PostgreSQL, pgvector, Qdrant, REST APIs, React, Next.js, Tailwind CSS
Infra & DevOps	Kubernetes, controller-runtime, Docker, Prometheus, OpenTelemetry, GitHub Actions (CI/CD), Git

ACHIEVEMENTS

- Secured **3rd place** for an AI-based Task Scheduler at a competitive technical event; participated in multiple hackathons.